



Physics-informed neural network with moving boundary constraints for modeling hydraulic fracturing

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ABSTRACT

Hydraulic fracturing is an essential technique for hydrocarbon extraction. While mathematical models are commonly utilized to simulate hydraulic fracture propagation, they are computationally expensive due to the complexities of moving boundary conditions. Recent advances in data-driven models have reduced the computational burden; however, these approaches often fail to preserve the physical interpretability. To address this limitation, a physics-informed neural network (PINN) has emerged as a promising methodology that integrates governing physics-based equations into a loss function. However, conventional PINNs still face challenges in accurately modeling the moving boundary nature of fracture propagation, particularly due to the evolving domain of interest over time. In this study, we propose a PINN with a moving boundary constraint (MB-PINN) for hydraulic fracture propagation. This approach employs a dual-network architecture to model the spatiotemporal evolution of fractures and enforce constraints on moving boundaries. Additionally, this architecture captures the nonlinear effects across diverse scenarios by incorporating fracturing conditions as input variables. Our framework improves the accuracy of fracture propagation modeling by reducing prediction errors compared to conventional PINNs, while significantly enhancing the computational efficiency compared to complex high-fidelity models. These advancements make it a scalable and reliable tool for simulating complex fracture dynamics under diverse operational conditions in real-world applications.

1. Introduction

Hydraulic fracturing is essential to enhance the extraction of hydrocarbon resources. The process begins with explosions inside the wellbore to create initial fractures. This is followed by the injection of a clean fluid under high pressure for fracture propagation. Later, the proppant is pumped to extend and widen the pathways that facilitate the flow of trapped oil and natural gas (Gu and Hoo, 2014; Adachi et al., 2007). In this case, the hydraulic fracture geometry significantly affects the permeability of the formation, thereby affecting the efficiency and profitability of the extraction process (Bello et al., 2022). Hence, accurate modeling techniques for the prediction and optimization of hydraulic fracture geometry are necessary.

The mathematical model of the hydraulic fracture geometry connects the fracture mechanics with the pressure distribution derived from fluid mechanics. Classical hydraulic fracturing models include the Perkins-Kern-Nordgren (PKN) model (Perkins and Kern, 1961; Nordgren, 1972), Kristianovich-Geertsma-de Klerk (KGD) model (Geertsma

and De Klerk, 1969), pseudo 3D (P3D) model (Simonson et al., 1978), and planar 3D (PL3D) model (Clifton and Abou-Sayed, 1979). These approaches solve nonlinear partial differential equations (PDE) with moving boundary conditions by utilizing numerical methods (Kovalyshen and Detournay, 2010; Lecampion et al., 2018; Chen et al., 2021; Bangi and Kwon, 2020). However, the inherent complexity of moving boundary conditions necessitates sophisticated meshing techniques and additional computational effort, which pose significant challenges for immediate prediction and optimization. Specifically, the moving boundary problem involves dynamic domain changes that require continuous mesh generation, which often leads to mesh distortion. Furthermore, maintaining numerical stability and ensuring convergence is increasingly challenging, as minor interpolation errors from mesh deformation can accumulate (Cai et al., 2017). These issues necessitate the implementation of advanced remeshing strategies and high-order numerical schemes, which increase the computational complexity (Anselmann and Bause, 2022).

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In recent years, surrogate models have been proposed to reduce the computational cost of hydraulic fracturing simulations. These include data-driven and reduced-order modeling approaches, such as local dynamic mode decomposition with control (DMDc) for model predictive control (Narasimam and Kwon, 2017; Bangi et al., 2019), local POD-Galerkin for nonlinear PDEs with time-dependent domains (Narasimam et al., 2017; Sidhu et al., 2018; Li et al., 2020), sparse regression via SINDy for interpretable reduced models (Narasimam and Kwon, 2018; Zalavadia and Sankaran, 2024), extended DMD (EDMD) leveraging Koopman operator theory (Narasimam and Kwon, 2020; Klie and Florez, 2019; Lin et al., 2022), and linear approximations for control applications (Siddhamshetty et al., 2018; Cardoso and Durlafsky, 2010). Building on these efforts, recent advances in general data-driven models have aimed to further reduce the computational burden of numerical models. Deep neural networks (DNNs), in particular, have emerged as promising tools due to their multilayer architecture and ability to approximate complex nonlinear functions. DNNs have been extensively applied across domains, such as wastewater treatment (Zhao et al., 2021), crystallization (Lu et al., 2024), porous media (Marcato et al., 2021, 2023), polymerization reactors (Zhong et al., 2019), and biomass pyrolysis (Lu et al., 2022a). However, despite their versatility, DNNs often require large, high-quality datasets and may struggle to preserve physical consistency, particularly in complex systems such as hydraulic fracturing. In such scenarios, limited or noisy data can hinder the ability of the model to capture the underlying physics, leading to unreliable predictions (Xue et al., 2021; Hou et al., 2024).

To address these limitations, a physics-informed neural network (PINN) was introduced by Raissi et al. (2019). The central concept of PINN involves integrating the residuals of the governing equations into the loss function, thereby embedding fundamental principles throughout the neural network. Although there have been other hybrid modeling-based approaches in the literature that integrate mechanistic models with data-driven models, they often operate by embedding data-driven components into the parameter space of the mechanistic system (Shah et al., 2022; Pahari et al., 2024; Shah et al., 2024; Sitapure and Kwon, 2023). This approach is computationally demanding, particularly when dealing with complex coupled PDEs or moving boundaries. In contrast, PINNs offer a more efficient alternative by directly embedding physical laws into the learning process. PINN has been utilized to solve complex problems in the areas of heat source systems (Liu et al., 2022), battery modeling and prognosis (Nascimento et al., 2021), cavitation flow (Ouyang et al., 2023), plasma simulation (Zhang et al., 2022), reactor control (Faria et al., 2024), optimization (Ryu et al., 2024), and transient flow of natural gas (Zhang and Shafieezadeh, 2023), while ensuring physical consistency. Despite these advancements, the implementation of PINN in problems involving moving boundary conditions remains challenging. Previous studies typically assumed that the rate of change of the moving boundary is known, simplifying the problem by positioning the collocation points at predefined interfaces. For instance, Lu et al. (2022b) defined an empirical loss function at moving boundaries to predict the hydraulic fracture geometry. Likewise, Zhu et al. (2024) refined the method by assigning fine collocation points at moving boundaries to solve incompressible flow problems. However, these approaches face significant challenges in practical applications. Without prior knowledge of the boundary evolution, it is difficult to accurately define the loss functions or impose appropriate constraints. This limitation highlights the need for adaptive methods that dynamically learn and adjust to changing boundaries without relying on predefined trajectories.

In this study, we developed a PINN with a moving boundary constraint (MB-PINN) for hydraulic fracturing. We utilized a dual network, as proposed by Wang and Perdikaris (2021) for moving boundary problems, to simultaneously determine the evolution of hydraulic fractures and the rate of change of the moving boundary. In this method, the moving boundary domain is inferred from the solution, and moving boundary constraints are iteratively imposed on the corresponding

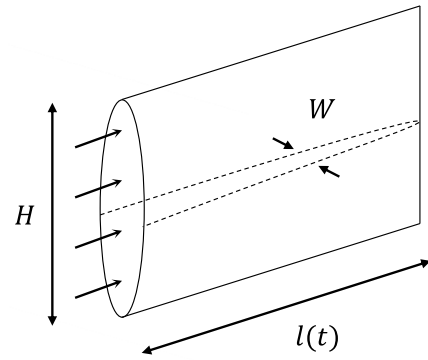


Fig. 1. Schematic of the PKN model.

domain. However, complex interactions between the two networks can lead to a reduction in convergence. Therefore, we included system-specific regularization terms in the loss function to ensure accurate identification of the time-dependent spatial domain. Moreover, the developed model incorporates fracturing conditions (e.g., fluid properties like viscosity and injection rate) as input variables and predicts the fracture width and length as output variables. Consequently, we demonstrated the effectiveness of the developed model by evaluating its accuracy using extensive validation data across 100 fracturing conditions, covering both interpolation and extrapolation ranges. Compared to existing approaches, our method demonstrated improved convergence and prediction accuracy, with enhancements of 35.2% in the interior regions and 73.6% at the moving boundary, by dynamically learning the evolution of the spatiotemporal domain without relying on prior assumptions regarding its trajectory. Additionally, we conducted a comprehensive global sensitivity analysis to quantify the influence of key input parameters on the fracture width and length, providing novel insights into parameter interactions that were not captured by previous models.

The remainder of this paper is organized as follows: Section 2 describes the mathematical representation of hydraulic fracture propagation. Section 3 introduces the methodology of the proposed model. Section 4 presents the modeling results and discussion. Finally, Section 5 provides concluding remarks.

2. Hydraulic fracture propagation

In this study, the mathematical formulation of the PKN model was utilized to define the geometry of a hydraulic fracture (Perkins and Kern, 1961). This model assumes that the fracture propagates with a constant height along its length, as shown in Fig. 1 (Adachi et al., 2007). Additional assumptions are as follows: (i) the length of the fracture is greater than its width; (ii) the fracture is confined to a single layer; and (iii) the rock properties are considered to be constant (Gu and Hoo, 2014). These assumptions simplify the process of hydraulic fracture propagation, leading to a significant reduction in computational demands.

The fluid momentum of hydraulic fracturing is described by the lubrication equation, given that the fracture width is substantially smaller than the fracture length (Pinkus et al., 1962). This equation enables efficient modeling of the fluid flow and pressure gradients:

$$\frac{dP}{dx} = -\frac{64\mu q_x}{\pi H w^3} \quad (1)$$

where P is the pressure, x is the spatial coordinate, μ is the fracturing fluid viscosity, q_x is the local flow rate in the x -direction, H is the fracture height, and w is the fracture width. Due to the large stress between the horizontal layer and the surrounding rock, the fracture remains

confined within a single layer, preventing vertical propagation. Under these conditions, Eq. (2) is derived to link the net pressure to the maximum fracture width, reflecting the planar geometry of the fracture:

$$w = \frac{2PH(1-\nu^2)}{E} \quad (2)$$

where ν is the Poisson's ratio, and E is the Young's modulus. In addition, implementing the local fluid mass balance to the fracturing fluid results in the following equation (Nordgren, 1972):

$$H \frac{\partial w}{\partial t} + \frac{\partial q_x}{\partial x} + HU = 0 \quad (3)$$

where t represents the injection time of the fracturing fluid, and U is the fluid leak-off rate. The fluid leak-off rate is described as follows:

$$U = \frac{2C_l}{\sqrt{t - \tau(x)}} \quad (4)$$

where C_l is the leak-off coefficient, and τ indicates the initiation leak-off time at the x location. Moreover, the initial and boundary conditions of Eq. (3) are defined as shown below:

$$w(z, 0) = 0 \quad (5)$$

$$q_x(0, t) = q_0 \quad (6)$$

$$w(l(t), t) = 0 \quad (7)$$

where q_0 denotes the rate of fracturing fluid at the wellbore, and $l(t)$ represents the fracture length as a function of injection time. When the fracture is initially closed, the fracture width can be defined by Eq. (5). Given that the injection rate of the fracturing fluid at the wellbore is fixed, the corresponding boundary condition is described by Eq. (6).

As the fracture propagates, the spatial domain changes over time. This dynamic behavior necessitates precise tracking of the domain boundary, which is governed by the moving boundary condition outlined in Eq. (7). Therefore, an accurate implementation of this equation is essential for surrogate modeling of hydraulic fracture propagation, particularly for PINN model discussed in Section 3.

3. Methodology

3.1. Overview of PINN

To illustrate the PINN framework, we consider the following equations:

$$\mathcal{F}(u, \nabla u, \nabla^2 u, \dots) = 0 \quad (8)$$

$$u(x, 0) - h(x) = 0 \quad (9)$$

$$u(x, t) - g(x, t) = 0 \quad (10)$$

where ∇ denotes differential operators, $u(x, t)$ is the solution, and $\mathcal{F}$ is the residual of the PDE, referring to Eq. (8). The variables x and t represent the spatial and temporal coordinates, respectively. The initial condition in Eq. (9) defines the initial state $u(x, 0)$ as $h(x)$ in the spatial domain. Meanwhile, the boundary condition in Eq. (10) specifies the behavior of $u(x, t)$ along the boundary domain as $g(x, t)$ over time.

We assume that the solution $u(x, t)$ can be approximated by a fully connected neural network $u_\theta(x, t)$, where θ indicates network parameters such as weights and biases. The parameter θ is optimized by minimizing the following loss function:

$$L(\theta) = \lambda_r L_r(\theta) + \lambda_{u_0} L_{u_0}(\theta) + \lambda_{u_b} L_{u_b}(\theta) + \lambda_u L_u(\theta) \quad (11)$$

where

$$\begin{aligned} L_u(\theta) &= \frac{1}{N_x N_t} \sum_{i=1}^{N_x} \sum_{j=1}^{N_t} |u_\theta(x_u^i, t_u^j) - u(x_u^i, t_u^j)|^2 \\ L_r(\theta) &= \frac{1}{N_r} \sum_{i=1}^{N_r} |\mathcal{F}(x_r^i, t_r^i)|^2 \\ L_{u_0}(\theta) &= \frac{1}{N_{u_0}} \sum_{i=1}^{N_{u_0}} |u_\theta(x_{u_0}^i, 0) - h(x_{u_0}^i)|^2 \\ L_{u_b}(\theta) &= \frac{1}{N_{u_b}} \sum_{i=1}^{N_{u_b}} |u_\theta(x_{u_b}^i, t_{u_b}^i) - g(x_{u_b}^i, t_{u_b}^i)|^2 \end{aligned} \quad (12)$$

Note that $L_u(\theta)$ quantifies the discrepancy between the observed and predicted values. The terms $L_r(\theta)$, $L_{u_0}(\theta)$, and $L_{u_b}(\theta)$ denote the residuals defined by Eqs. (8), (9), and (10), respectively. Additionally, the parameters $\{\lambda_u, \lambda_r, \lambda_{u_0}, \lambda_{u_b}\}$ serve as weight coefficients for each corresponding loss term, allowing for the adjustment of their relative importance in the overall training process.

3.2. PINN for hydraulic fracture propagation

Since the conventional PINN utilizes a single network to update the model parameters, it cannot accurately capture moving boundary conditions unless the trajectory is predefined. To incorporate the physical constraints related to the moving boundary conditions, we formulated the governing equations for hydraulic fracture propagation based on Eqs. (3), (5), and (6) as follows:

$$\frac{\pi H}{4} \frac{\partial w}{\partial t} - \frac{\pi E}{128\mu(1-\nu^2)} \left[3w^2 \left(\frac{\partial w}{\partial x} \right)^2 + w^3 \frac{\partial^2 w}{\partial x^2} \right] + HU = 0 \quad (13)$$

$$w(x, 0) = 0 \quad (14)$$

$$\frac{\pi E w^3}{128\mu(1-\nu^2)} \frac{\partial w}{\partial x} + q_0 = 0 \quad (15)$$

where the solution $w(x, t)$ represents the fracture width. We consider the following moving boundary condition according to Eq. (7).

$$w(l(t), t) = 0 \quad (16)$$

where $l(t)$ denotes the position of the fracture tip, defined by the fracture length as a function of time t .

To address the simultaneous learning of the unknown solution $w(x, t)$ and moving boundary $l(t)$, we propose a novel approach called PINN with Moving Boundary Constraint (MB-PINN). This framework utilizes two distinct neural networks: $w_\theta(x, t)$ for modeling the spatiotemporal solution and $l_\beta(t)$ for predicting the dynamic position of the moving boundary. Here, θ and β are independent model parameters, as depicted in Fig. 2. These networks interact during training by dynamically defining a moving boundary, thereby enabling the enforcement of moving boundary conditions without relying on predefined boundary locations.

In this setup, the network w_θ focuses on learning the solution governed by the physical laws of hydraulic fracturing, incorporating fracturing conditions $\mathcal{P}$ (such as fluid viscosity μ and injection rate q_0) as the input variables. Its loss function combines the empirical loss with the physical loss derived from the residuals of the PDEs, initial conditions, and boundary conditions, thereby ensuring accurate prediction of the fracture width. In contrast, network l_β explicitly models the temporal evolution of the fracture length. As expressed in the initial condition given by Eq. (17), the fracture length is constrained to be zero at the beginning of the hydraulic fracturing process, thereby effectively capturing the dynamics of the moving boundary.

$$l(0) = 0 \quad (17)$$

The networks w_θ and l_β are integrated into a unified framework, as their outputs jointly contribute to the moving boundary constraint.

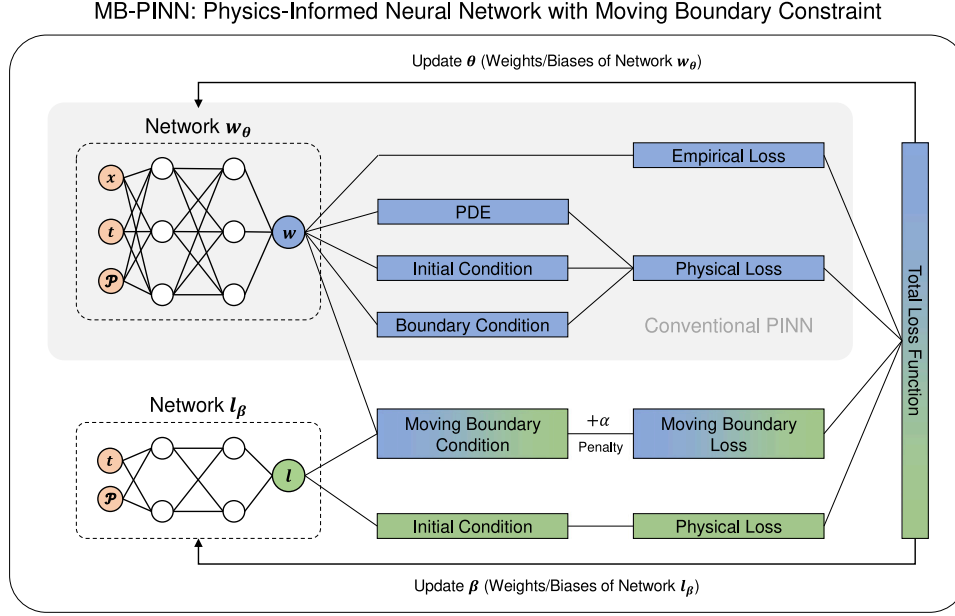


Fig. 2. Schematic illustration of MB-PINN for hydraulic fracture propagation. Conventional PINN utilizes only the network w_θ , which incorporates general physical constraints but excludes the moving boundary constraint. In contrast, MB-PINN employs two coupled networks: w_θ predicts the spatiotemporal fracture width, while l_β predicts the fracture tip position over time. These networks are indirectly coupled through total loss function, ensuring physical consistency between the fracture width and the moving boundary.

Specifically, network l_β identifies the moving boundary based on the output of the network w_θ . Conversely, the constraint derived from Eq. (16) is imposed beyond the corresponding boundary to ensure that the fracture width approaches zero.

In addition, we introduce a regularization term for network l_β to accurately identify the time-dependent spatial domain. This regularization term α is defined by the following equation to ensure model convergence during training:

$$\alpha = |l_\beta(t)|^2 + |l_\beta(t_f) - l(t_f)|^2 \quad (18)$$

where t_f represents the final time of the injection process. The components of Eq. (18) addresses the specific challenges as follows:

1. Stability of Moving Boundary: The network l_β models the moving boundary by tracking the fracture tip, where the fracture width approaches zero, which makes stable prediction challenging. To mitigate this issue, the regularization term $|l_\beta(t)|^2$ is introduced to specify the moving boundary to an initial point where the fracture width is zero.

2. Convergence at Termination: The term $|l_\beta(t_f) - l(t_f)|^2$ resolves the problem of predicting abrupt changes at the fracture tip. By incorporating the discrepancy between the predicted and actual fracture lengths at $t = t_f$, this regularization term enhances the model convergence and reduces the prediction errors at the final time of the hydraulic fracturing process.

Consequently, MB-PINN was trained to minimize the following loss function, which incorporates the overall constraints associated with hydraulic fracture propagation:

$$L(\theta, \beta) = \lambda_{s_0} L_{s_0}(\theta, \beta) + \lambda_{s_b} L_{s_b}(\beta) + \lambda_r L_r(\theta) + \lambda_{u_0} L_{u_0}(\theta) + \lambda_{u_b} L_{u_b}(\theta) + \lambda_u L_u(\theta) \quad (19)$$

where

$$\begin{aligned} L_{s_0}(\beta) &= \frac{1}{N_{s_0}} \sum_{i=1}^{N_{s_0}} |l_\beta(0)|^2 \\ L_{s_b}(\theta, \beta) &= \frac{1}{N_{s_b}} \sum_{i=1}^{N_{s_b}} |w_\theta(l_\beta(t^i), t^i)|^2 + \alpha \\ L_u(\theta) &= \frac{1}{N_x N_t} \sum_{i=1}^{N_x} \sum_{j=1}^{N_t} |w_\theta(x^i, t^j) - w(x^i, t^j)|^2 \\ L_{u_0}(\theta) &= \frac{1}{N_{u_0}} \sum_{i=1}^{N_{u_0}} |w_\theta(x^i, 0)|^2 \\ L_{u_b}(\theta) &= \frac{1}{N_{u_b}} \sum_{i=1}^{N_{u_b}} \left| \frac{\pi E (w_\theta^i)^3}{128 \mu (1 - \nu^2)} \frac{\partial w_\theta^i}{\partial x} + q_0 \right|^2 \\ L_r(\theta) &= \frac{1}{N_r} \sum_{i=1}^{N_r} \left| \frac{\pi H}{4} \frac{\partial w_\theta^i}{\partial t} - \frac{\pi E}{128 \mu (1 - \nu^2)} [3 w_\theta^{i^2} (\frac{\partial w_\theta^i}{\partial x})^2 + w_\theta^{i^3} \frac{\partial^2 w_\theta^i}{\partial x^2}] + H U \right|^2 \end{aligned} \quad (20)$$

where the collocation points for $L_r(\theta)$, $L_{u_0}(\theta)$, $L_{u_b}(\theta)$, and $L_{s_0}(\beta)$ are randomly sampled within their respective domains. These collocation points represent locations where the mean squared error of the fracture width and residuals of the governing equations were computed during the training process. Since the domain of the moving boundary is unknown, we initially sampled collocation points across the entire domain for $L_{u_b}(\theta, \beta)$. We then focused on the domain corresponding to the moving boundary by restricting the predictions. Based on these collocation points, the residual terms $L_r(\theta)$, $L_{u_0}(\theta)$, and $L_{u_b}(\theta)$ are defined according to Eqs. (13), (14), and (15), respectively. Note that $L_u(\theta)$ quantifies the discrepancies between the observed and predicted values. The exact solution for $L_u(\theta)$ was generated based on a numerical solver with the parameters listed in Table 1. Compared to the conventional PINN structure described in Section 3.1, the additional terms $L_{s_b}(\theta, \beta)$ and $L_{s_0}(\beta)$ are introduced to constrain the model, as defined by Eqs. (16) and (17), respectively.

Fully connected neural networks were utilized to approximate the solution of networks w_θ and l_β , each composed of multiple hidden layers with 100 units. After training, the two networks operate jointly

Table 1
Model parameters for hydraulic fracture propagation.

Description	Symbol	Unit	Value	Range
Fracture height	H	m	60	–
Poisson's ratio	ν	–	0.2	–
Young's modulus	E	Pa	4.2×10^5	–
Leak-off coefficient	C_l	$\text{m/s}^{1/2}$	6.0×10^{-5}	–
Fluid viscosity	μ	Pa s	–	[0.0001, 0.0009]
Fluid injection rate	q_0	m^3/s	–	[0.03, 0.07]

Table 2
Optimal hyperparameters of MB-PINN for hydraulic fracture propagation.

Description	Network w_θ	Network l_β
Iteration	10,000	10,000
Batch size	64	64
Optimizer	Adam	Adam
Learning rate	10^{-4}	10^{-4}
Activation function	Tanh	Sigmoid
Weight initialization	Glorot	Glorot
Number of hidden units	100	100
Number of hidden layers	3	2

during inference: l_β predicts the temporal evolution of the fracture tip, defining a moving boundary, and w_θ predicts the fracture width within this evolving spatial domain. Both networks were trained using the Adam optimizer with a learning rate of 10^{-4} . However, distinct activation functions, i.e., Tanh for network w_θ and sigmoid for network l_β , were chosen to constrain each solution within specific ranges. The hyperparameters for both networks are listed in Table 2. These parameters were identified using a grid search algorithm and remained constant throughout the study. Moreover, the training process was conducted on Intel® Xeon Gold 6226R (2.90 GHz, 16 core, supports up to 1 TB RAM) with NVIDIA® GeForce RTX 3090 GPU.

4. Results

In Section 4.1, we validate the proposed MB-PINN model using NN as the primary baseline model to demonstrate the effectiveness of incorporating complete physical constraints. The baseline model relies solely on the network w_θ without physical constraints. Furthermore, a predictive analysis was conducted in Section 4.2 to demonstrate the extrapolation capabilities of the injection time and fracturing conditions. In Section 4.3, a global sensitivity analysis is performed to evaluate the effect of fracturing conditions on fracture behavior. Finally, a comparison of the computational costs between MB-PINN and high-fidelity model was conducted.

4.1. Model validation

To demonstrate the overall performance of the proposed model, we compared the validation losses between MB-PINN and NN, as depicted in Fig. 3. Fig. 3(a), Fig. 3(b), Fig. 3(c), and Fig. 3(d) represent the histories of $L_u(\theta)$, $L_r(\theta)$, $L_{u_0}(\theta)$, and $L_{u_b}(\theta)$, respectively. In the initial stages of training, NN exhibits rapid convergence due to its lack of physical constraints. However, this purely data-driven approach can increase the risk of overfitting, leading to poor generalization of unseen data. In contrast, by the end of the training, MB-PINN consistently showed lower loss values than NN across all metrics, demonstrating that the integration of physical constraints enhances accuracy and convergence. The results also highlight the effective balance across all loss functions, with a steady reduction in losses observed for MB-PINN.

The final loss for the initial condition shows the smallest difference between MB-PINN and NN, with values of 4.39×10^{-4} and 4.44×10^{-4} , as illustrated in Fig. 3(c). This is due to the simplicity of the initial condition, which both models can adapt to easily. On the other hand, for the boundary condition in Fig. 3(d), MB-PINN achieves a

significantly lower final loss of 1.37×10^{-3} compared to 6.08×10^{-3} for NN, indicating its superior ability to capture the rate of change of the moving boundary. This substantial difference highlights the effectiveness of MB-PINN in incorporating moving boundary constraints, which are critical for accurate predictions and enhanced convergence.

Following this initial analysis, we assessed the accuracy of each model in terms of fracture width. Fig. 4(a) and 4(b) compare the fracture width predictions of MB-PINN and NN with the exact solutions. NN displayed noticeable discrepancies in Fig. 4(b), particularly near the fracture tip, where the fracture width decreases steeply to zero. This is because of the inherent limitations of DNNs in handling sharp gradients or abrupt changes (Vahab et al., 2023), which require explicit physical constraints for accurate modeling. In contrast, as illustrated in Fig. 4(a), MB-PINN demonstrates a close alignment with the exact solutions, even at the fracture tip. In regions excluding the fracture tip, the mean squared error for MB-PINN was 8.36×10^{-10} , whereas that for NN was 1.29×10^{-9} . This comparison suggests that physical constraints contribute to the overall accuracy of MB-PINN. Additionally, the error at the fracture tip shows a substantial difference, with MB-PINN achieving an error of 5.28×10^{-8} compared to 2.00×10^{-7} for NN, highlighting the significant impact of the moving boundary constraint.

To verify the significance of the moving boundary constraint on the predictive performance, we further evaluated the model accuracy by calculating the mean absolute error (MAE) of the fracture length across MB-PINN, PINN, and NN models. In this case, PINN utilizes only the network w_θ with general physical constraints, excluding the moving boundary constraint. Given that both PINN and NN rely solely on the network w_θ , the fracture length is inferred indirectly based on the prediction of the fracture width. As shown in Fig. 4(c), the performance of the models shows a clear trend, with MB-PINN consistently outperforming both PINN and NN over the entire range of injection times. Notably, the comparison between MB-PINN and PINN highlights the effectiveness of the moving boundary constraint, which improves the model's ability to capture the complex dynamics at moving boundaries. In addition, MB-PINN and PINN maintained a lower MAE in regions where NN exhibited a substantial increase, particularly at approximately 50 and 500 s of injection time. This is primarily because during the early stages of injection, the abrupt fluid entry into the wellbore creates steep pressure gradients, making it difficult to capture the flow dynamics. In the later stages, fluid leak-off into the surrounding formations reduces the flow to the fracture tip, causing localized flow variations that further complicate the modeling. By embedding general physical constraints, such as PDEs, initial conditions, and boundary conditions, MB-PINN and PINN can handle such complexities more effectively, resulting in improved accuracy across the critical phases of the injection process.

4.2. Analysis of the model's extrapolation capability

We demonstrated the extrapolation capability beyond the training regime for both the injection time and fracturing conditions in Fig. 5. As shown in Fig. 5(a), the MAE of the fracture width was analyzed to compare the performance of MB-PINN and NN across untrained injection times. Note that the injection time within the training range spans from 0 to 500 s, with each line corresponding to the average MAE across 36 operating conditions and shaded regions indicating the minimum and maximum values. Both MB-PINN and NN showed significantly higher errors in the extrapolation regime than in the interpolation regime. However, MB-PINN consistently outperformed NN across the entire extrapolation regime, with notably smaller deviations. This result suggests that MB-PINN provides more generalized predictions, demonstrating superior extrapolation capability over NN.

In contrast, the MAE of the fracture width was evaluated under various fracturing conditions by varying the viscosity and injection rate of the fracturing fluid. Fig. 5(b) and Fig. 5(c) display the contour plots of the MAE for MB-PINN and NN, respectively. The area within the

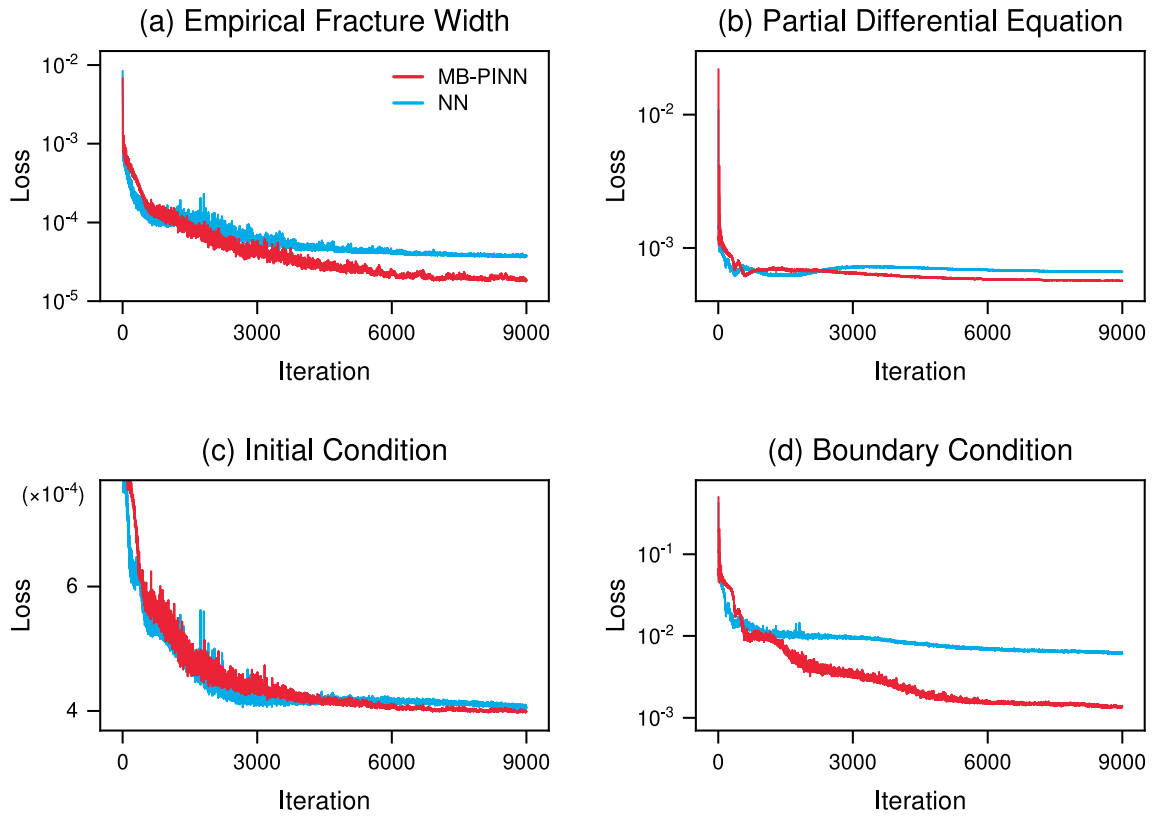


Fig. 3. Training curves showing the validation loss of MB-PINN and NN for (a) empirical fracture width, (b) PDE, (c) initial condition, and (d) boundary condition.

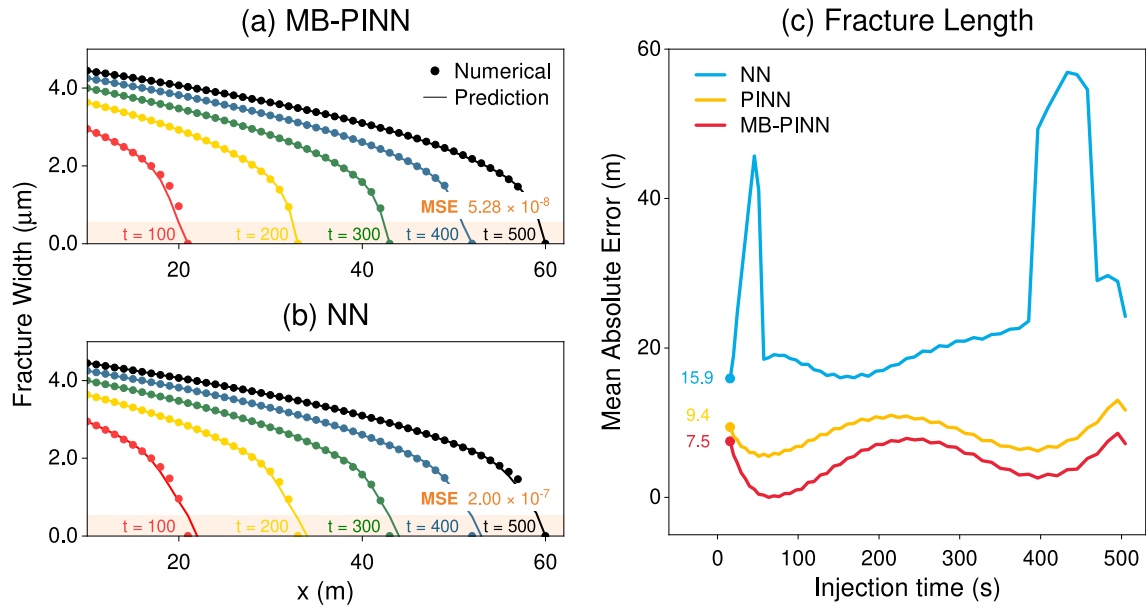


Fig. 4. Performance comparison between MB-PINN, PINN, and NN: (a) Fracture width predicted by MB-PINN and the exact solution, showing strong agreement even near the fracture tip; (b) fracture width predicted by NN and the exact solution, revealing noticeable errors near the fracture tip; and (c) MAE of fracture length over injection time, demonstrating superiority of MB-PINN during the early and late injection phases. The highlighted regions in (a) and (b) indicate the fracture tip area where the moving boundary constraint $w(l(t), t) = 0$ is enforced, representing the most challenging region for accurate prediction due to sharp gradients.

white box represents the training range, while the outside corresponds to the untrained range. MB-PINN exhibited significantly lower errors than NN, particularly in the extrapolation regime. The prediction accuracy declines at lower injection rates and higher viscosities because of transient flow effects arising from reduced momentum and increased resistance in the fracturing fluid. However, MB-PINN can capture these

complex dynamics by adhering to the governing equations, ensuring physical consistency. In contrast, NN is prone to overfitting within the training range, limiting its accuracy to localized conditions.

We further demonstrated the extrapolation capabilities of the proposed model by evaluating the net pressure under new fracturing conditions (i.e., for the purpose of testing extrapolation capabilities).

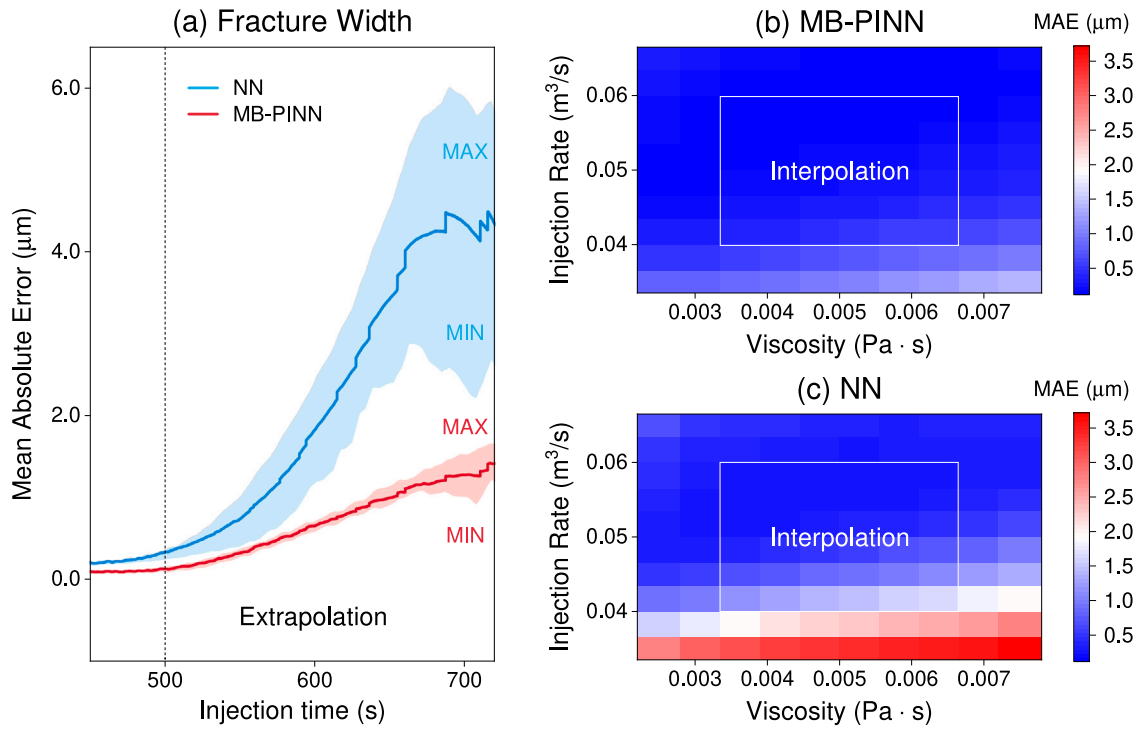


Fig. 5. Extrapolation performance comparison between MB-PINN and NN: (a) MAE of fracture width over untrained injection times, with MB-PINN showing lower error and variance than NN beyond the training range; (b) contour plot of fracture width MAE for MB-PINN across varying fluid viscosities and injection rates, demonstrating robust performance in both trained and untrained regimes; and (c) contour plot of fracture width MAE for NN, indicating higher errors in the extrapolation regime.

Net pressure, defined as the difference between the fracturing fluid pressure and closure pressure (Hoss et al., 2017), is influenced by several parameters, including the viscosity and injection rate of the fracturing fluid. Thus, an accurate prediction of the net pressure is crucial for ensuring proper fracture propagation and minimizing the risk of uncontrolled scenarios.

In Fig. 6, the actual values from the numerical solvers and predicted values from MB-PINN were compared over the injection time. For the PKN model, the net pressure rises sharply as the fracture forms, resulting in a steep pressure gradient due to the fluid being injected into a narrow fracture. As the fracture gradually expands, the growing fracture volume reduces the pressure gradient. Eventually, the net pressure stabilizes as further fluid injection no longer significantly increases the fracture size. MB-PINN predictions closely followed this trend, effectively capturing the dynamics of the net pressure during hydraulic fracture propagation.

Fig. 6(a) illustrates the net pressure at the wellbore under various viscosity conditions. As the viscosity of the fracturing fluid increases from 0.0025 Pa s to 0.0075 Pa s, the net pressure rises correspondingly due to the increased resistance within the fracture, leading to a delay in the injection time required for the fracture to reach a specific point. Conversely, Fig. 6(b) depicts the net pressure at the wellbore under different injection rates. As the injection rate increases from 0.035 m³/s to 0.065 m³/s, the net pressure similarly rises due to the greater volume of the fracturing fluid. Notably, MB-PINN accurately predicted the net pressure across both the extrapolated fracturing conditions and injection times, demonstrating its extrapolation capability to capture the underlying nonlinear relationship between these factors.

4.3. Sensitivity analysis

Fig. 7 provides a comparative analysis of the variance-based sensitivity between MB-PINN and the high-fidelity model, evaluating the effect of fracturing conditions on hydraulic fracture propagation. To measure the feature importance across the fracturing conditions, we

conducted a global sensitivity analysis by evaluating the Sobol Index (Sobol, 2001), which decomposes the output variance into fractions attributed to each input variable.

In Fig. 7(a), we computed the Sobol total sensitivity index for two key fracturing conditions — viscosity and injection rate — on the fracture length. Both MB-PINN and the high-fidelity model showed that the injection rate had a greater influence on the fracture length than viscosity, with Sobol sensitivity indices of approximately 0.6 for the injection rate and approximately 0.41–0.42 for viscosity. This similarity in the Sobol sensitivity indices between the two methods demonstrates MB-PINN's capability of MB-PINN to capture the dominant factors with high fidelity.

Fig. 7(b) compares the computational costs of MB-PINN and the high-fidelity model. MB-PINN demonstrated significantly lower computation times, averaging approximately 0.025 s, compared to nearly 7.33 s for the high-fidelity model. Additionally, the high-fidelity model exhibited greater variability in computation time across fracturing conditions, with a variance of 2.17×10^{-1} . In contrast, MB-PINN maintains both a low computational cost and minimal variation, with a variance of 7.00×10^{-6} , making it suitable for dynamic sensitivity analysis. As shown in Fig. 7(c), the reduced computational cost enables MB-PINN to effectively capture the dynamic changes in the sensitivity indices.

In Fig. 7(c), the Sobol total sensitivity indices of the injection rate and viscosity were evaluated over the injection time. Initially, the injection rate exhibits the highest sensitivity because it directly influences the steep pressure gradient. As injection progresses, the sensitivity to injection rate decreases slightly, while the sensitivity to viscosity increases. This shift indicates that viscosity becomes more important in controlling fluid flow as the fracture stabilizes. In the later stages, the sensitivity to the injection rate increases again, as maintaining sufficient pressure to deliver fluid to the fracture tip becomes crucial. Despite these variations, the injection rate consistently maintained a higher contribution throughout the process, highlighting its dominant role in determining the fracture length.

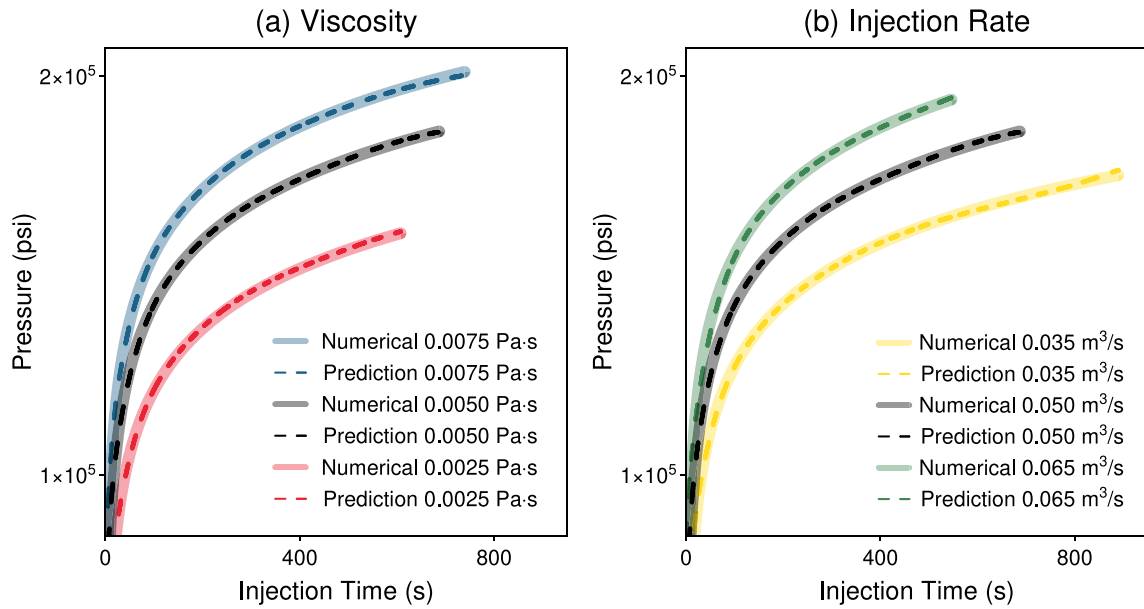


Fig. 6. Comparison of net pressure at the wellbore between predictions and numerical solutions: (a) Net pressure over time under varying fluid viscosities, showing increased pressure with higher viscosity; and (b) net pressure over time under varying injection rates, with higher injection rates yielding greater pressure.

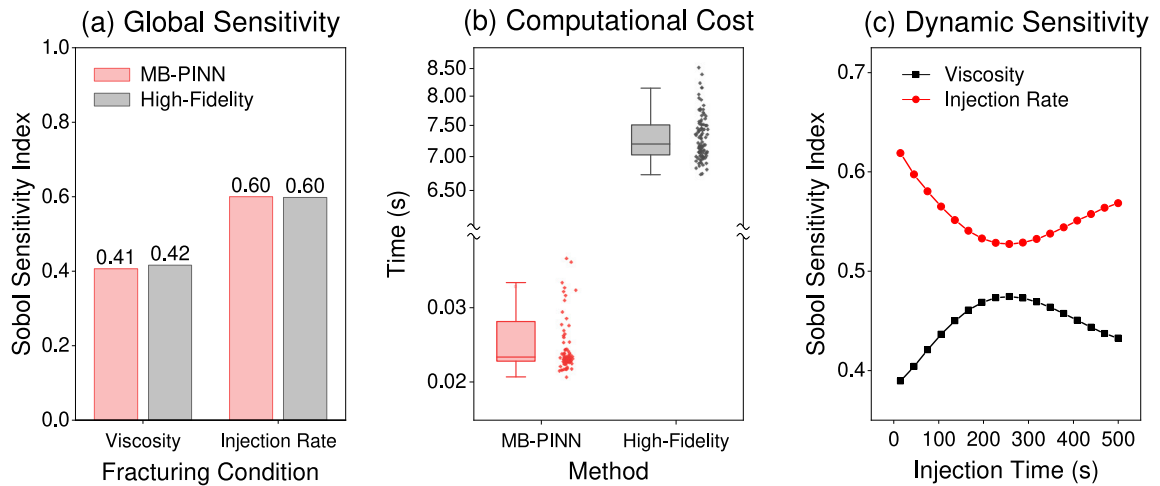


Fig. 7. Comparison of sensitivity analysis results and computational efficiency between MB-PINN and high-fidelity model: (a) Sobol total sensitivity indices of fracture length with respect to viscosity and injection rate, showing consistent dominance of injection rate; (b) comparison of computational cost across fracturing conditions, with MB-PINN achieving lower runtime; and (c) dynamic sensitivity indices over injection time, illustrating the shifting influence of viscosity and injection rate during different stages of fracture propagation.

5. Conclusion

This study presents a physics-informed neural network with a moving boundary constraint (MB-PINN) for modeling hydraulic fracture propagation. While inspired by the general architecture proposed by Wang and Perdikaris (2021), our dual-network formulation is specifically designed to address the challenges unique to hydraulic fracturing, including moving boundaries and parametric dependencies. By leveraging a dual-network architecture, MB-PINN simultaneously captures the spatiotemporal dynamics of the fracture width and the evolution of the moving boundary. Physical constraints, including governing equations and fracturing conditions, such as fluid viscosity and injection rate, were embedded into the training process to improve accuracy and stability. Compared with conventional neural networks and the baseline PINN, MB-PINN exhibited superior predictive accuracy, particularly near the fracture tip, and robust extrapolation performance across untrained conditions. Furthermore, it achieves a significant reduction

in computational cost while maintaining fidelity, making it a promising surrogate model for real-time simulations and sensitivity analyses. Future research will focus on scaling this framework to 3D and multiphysics systems and integrating it with hybrid approaches that combine data-driven models with traditional first principles to enhance its robustness and applicability in complex engineering domains.

CRediT authorship contribution statement

Yubin Ryu: Writing – original draft, Validation, Methodology, Writing – review & editing, Visualization, Software, Investigation, Data curation, Formal analysis, Conceptualization. **Parth Shah:** Writing – review & editing, Supervision, Investigation, Conceptualization, Writing – original draft, Software, Formal analysis. **Jonggeol Na:** Writing – review & editing, Supervision, Methodology, Conceptualization, Writing – original draft, Project administration, Formal analysis. **Joseph Sang-il Kwon:** Writing – original draft, Project administration, Funding

acquisition, Conceptualization, Writing – review & editing, Supervision, Methodology, Formal analysis.

Code availability

The code for the modeling is available at <https://github.com/YubinRyu/MB-PINN>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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